Experiment: Installation and Configuration of MongoDB

Aim

To install and configure MongoDB on a computer and understand its basic setup and working.

Theory

MongoDB is an open-source, NoSQL database management system used to store and manage large amounts of data. Unlike traditional relational databases, which store data in tables and rows, MongoDB stores data in flexible, JSON-like documents called BSON (Binary JSON). These documents are organized into collections.

MongoDB is useful for applications that require flexibility, scalability, and fast data access. It allows developers to store structured, semi-structured, and unstructured data without requiring a fixed table structure.

Features of MongoDB

1. Document-Oriented: Stores data in the form of documents containing key-value pairs.
2. Schema Flexibility: Documents in the same collection can have different structures.
3. Scalability: Supports horizontal scaling to manage large amounts of data.
4. High Performance: Provides efficient data storage and retrieval.
5. Indexing: Supports indexes to improve query performance.
6. Replication: Maintains multiple copies of data to improve availability and reliability.

Installation and Configuration of MongoDB

MongoDB can be installed and configured by following these steps:

1. Visit the official MongoDB website: https://www.mongodb.com/try/download/community
2. Select the appropriate operating system, version, and package format.
3. Download and run the MongoDB Community Server installer.

4. Follow the installation instructions and select the Complete installation option.
5. Configure MongoDB as a Windows service so that it can start automatically.
6. Install MongoDB Compass, a graphical user interface (GUI) used to view and manage databases.
7. After installation, start the MongoDB service.
8. Open MongoDB Compass and connect using the default connection string:

mongodb://localhost:27017

Basic MongoDB Commands

MongoDB provides commands to perform database operations.

Command	Purpose
show dbs	Displays all databases
use student_db	Creates or switches to a database
db	Displays the current database
db.createCollection("students")	Creates a collection
show collections	Displays all collections
db.students.insertOne({name:"Rahul", age:21})	Inserts a document
db.students.find()	Displays documents
db.students.deleteOne({name:"Rahul"})	Deletes a matching document

Advantages of MongoDB

- Easy to install and configure.
- Stores data in a flexible document format.
- Handles large volumes of data efficiently.
- Supports fast queries and indexing.

- Suitable for web applications, mobile applications, and real-time systems.

Conclusion

MongoDB was successfully installed and configured. The basic database operations, such as creating a database, creating collections, inserting documents, and retrieving data, can be performed using MongoDB commands. It provides a flexible and efficient way to store and manage data compared to traditional relational databases.